

NAME: _____

PERIOD: _____

DATE: _____

Two Test Plans for Cure Time

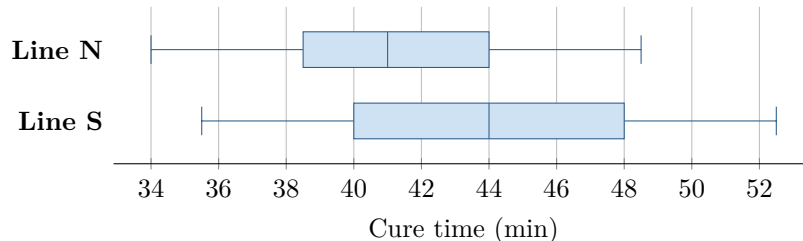
AP Statistics · Unit 4 · Topic 4.9 · B: 2027-01-28 · E: 2027-03-17

[STAR] TODAY'S PROBLEM

Suppose existing lines N (new formula) and S (standard) make the same sealant. Independently, a team samples 18 of 600 recent N batches and 18 of 540 recent S batches. Cure-time summaries are $\bar{x}_N = 41.2$, $s_N = 3.6$, $\bar{x}_S = 44.0$, and $s_S = 4.1$; boxplots are shown. Before sampling it asked, “Do the two populations differ in mean cure time?” Formula was not assigned to lines.

Rhea: “Preserve the pre-data direction, use population means, and separate association from causation.”

Sol: “Since $41.2 < 44.0$, use a lower tail with sample means; significance proves formula caused faster curing.”



FIRST TAKE — before the video (5 min)

Which parts of each plan do the question's timing and the design support? Name one detail to verify.

[BOOK] LET THE QUESTION—NOT THE SAMPLE—SET THE TAIL (keep on the page)

For two independent groups with a quantitative response, use a two-sample t test for $\mu_1 - \mu_2$. The null states no population difference: $H_0 : \mu_1 - \mu_2 = 0$. Choose $<$ or $>$ only when the research question specified that direction before seeing the data; otherwise use $\neq$. Hypotheses use population parameters, never observed sample means. Check independent random samples or random assignment, both 10 percent conditions when sampling without replacement, and both sample shapes when either $n < 30$. Random sampling supports population scope; random assignment is needed to isolate a causal treatment effect.

Finish after the video:

DOK 1 (a) Identify the response, both samples, and $\bar{x}_N - \bar{x}_S$. Did the pre-data question specify a direction?

DOK 2 (b) Name the procedure; define both parameters; state H_0 in N-minus-S order; and verify independence, 10 percent, and shape.

DOK 3 (c) Adjudicate both plans. Defend the alternative and notation, diagnose selecting a tail from the observed order, and bound a significant result. Explain the misleading belief invited by each rejected choice.

Frame: The pre-data wording calls for H_a : _____ because _____.

Frame: A significant result could support _____, but not _____, because _____.

EXIT — turn this in

One thing the video changed about my first take: _____